Student Dropout Prediction: A Machine Learning Analysis

DSAA2011(L01) Course Project - HKUST(GZ) 2026 Spring

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Date: 7 May 2026

Introduction

This study is based on the "Student Dropout and Academic Success" dataset from the UCI repository. Through a complete machine learning pipeline, we aim to predict students' academic outcomes.This presentation will focus on the following two parts:

1. **Clustering Analysis:** We will present the clustering analysis, detailing the process from *algorithm selection* (addressing the limitations of hard boundaries $\rightarrow$ choosing GMM), *parameter tuning*, to the *validation of clustering result reasonableness*.
2. **Open-ended Exploration:** We will elaborate on the open exploration conducted to enhance predictive performance.

Clustering (1/2): Algorithm Selection & Tuning

1. Hard-Boundary Models

- **Exploration:** Tested Hierarchical (Ward) and K-Means.
- **Optimal K:** $K = 4$ provided the best granularity.
- **Limitation:** While K-Means performed well, rigid spherical boundaries are less flexible in handling the severe structural overlap of the "Enrolled" class.

$\Rightarrow$ *Conclusion: A probabilistic approach is needed to handle the overlap.*

2. Final Selection: GMM

- **Best Config:** $K = 4$ with **Tied Covariance**.
- **Why GMM?** Soft-assignments handled overlap naturally; "Tied" covariance reduced 40D overfitting.

Performance Comparison ($K = 4$)

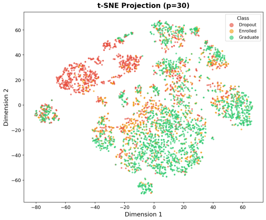
Algorithm	ARI	NMI
Hierarchical (Ward)	0.153	0.149
K-Means	0.313	0.230
GMM (Tied)	**0.334**	**0.273**

Clustering (2/2): Core Insight & Visualizations

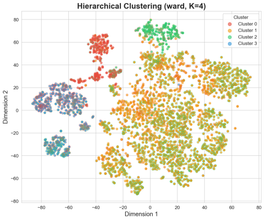
Insight: Natural Label Consistency

- **Consistency:** Clustering results seem to be basically consistent with true labels.
- **Ablation Check:** Removing target-encoded features yielded identical structures.

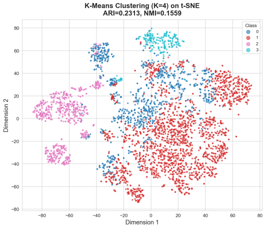
Original Labels



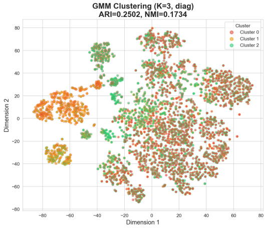
Hierarchical (Ward)



K-Means ($K = 4$)



GMM (Tied)



Part C

Part D

Reference

Realhino, V., et al. (2022). Student Dropout and Academic Success. UCI Machine Learning Repository. https://doi.org/10.24432/C5MC89

Q & A

Thank You!